

BUSINESS INSIGHTS REPORT

Supply Chain Analysis

₹5,77,605 Total Revenue	100 Unique SKUs	5 Total Suppliers	46,099 Products Sold	₹49.46 Avg Price
45.54% Avg Defect Rate	17% Delayed Shipments	53% Low Stock Products	36% Failed Inspection	₹554.81 Total Shipping Cost

KEY FINDINGS:

1. Skincare Dominates Revenue, But All Categories Are Competitive

Skincare generates maximum revenue of INR 2,41,628 (41.8% of total revenue), followed by Haircare at INR 1,74,455 (30.2%) and Cosmetics at INR 1,61,521 (28.0%). Revenue split shows healthy category mix but Skincare dominance implies higher concentration risk on demand. The top 5 revenue-generating SKUs are: SKU51 (INR 9,866), SKU38 (INR 9,692), SKU31 (INR 9,655), SKU90 (INR 9,593), and SKU2 (INR 9,578).

Product Type	Total Revenue (INR)	% of Total	Revenue Rank
Skincare	2,41,628	41.8%	#1
Haircare	1,74,455	30.2%	#2
Cosmetics	1,61,521	28.0%	#3

2. Nearly Half of Products Have Defects (45.54% Defect Rate)

The average defect rate across all 100 products is 45.54%, nearly 9x the acceptable industry threshold of 5%. 91 out of 100 products exceed the 5% defect threshold, and 45 products are formally flagged as 'High Defect'. Supplier 5 has the worst average defect rate at 53.3%, while Supplier 1 is the best performer at 36.1% (still far above acceptable). 36% of products outright failed quality inspection, and only 23% passed. This represents a systemic quality failure requiring immediate supplier audits.

Supplier	Products	Avg Defect Rate	Avg Lead Time (days)	Total Revenue (INR)	Risk Level
Supplier 5	18	53.3%	14.7 days	1,10,343	HIGH RISK
Supplier 3	15	49.3%	14.3 days	97,796	MEDIUM RISK
Supplier 2	22	47.3%	16.2 days	1,25,467	MEDIUM RISK
Supplier 4	18	46.7%	17.0 days	86,469	MEDIUM RISK
Supplier 1	27	36.1%	16.8 days	1,57,529	LOW RISK

Note: Risk level based on Q50 Supplier Health Report query. Defect rate thresholds: >50% = HIGH RISK, 40-50% = MEDIUM RISK, <40% = LOW RISK.

3. Shipping Delay Analysis: 17% Orders Delayed, All Carriers Show Negative Avg Delay

17 out of 100 shipments (17%) arrive after their expected lead time. Interestingly, when looking at average delay days per carrier, all three carriers (A, B, C) show negative average delay values (-8.82, -10.60, -10.97 days), meaning most shipments actually arrive EARLY on average — the 17 delayed ones represent outlier cases where specific routes or suppliers cause significant late deliveries. Route C has the highest average shipping cost (INR 5.90/unit), followed by Route B (INR 5.55) and Route A (INR 5.38). Road and Air transportation modes have the highest cost-to-revenue ratio at 0.10%, while Rail and Sea are most cost-efficient at 0.09%.

Shipping Carrier	Avg Delay (days)	Status	Delayed Orders	Performance
Carrier A	-8.82 days	Early avg.	Some outliers	Best
Carrier B	-10.60 days	Early avg.	Some outliers	Good
Carrier C	-10.97 days	Early avg.	Some outliers	Good

4. Critical Stockout Risk: 53% Products on Low Stock, Including Top Revenue Earners

53 out of 100 products have stock levels below 50 units, classified as 'Low Stock'. Only 3 products are 'Well Stocked' (>100 units). More critically, 30 high-revenue products (revenue above average of INR 5,776) are simultaneously in low stock creating an immediate stockout risk for the company's best earners. The worst cases include SKU31 (Skincare, stock=6 units, revenue INR 9,655), SKU2 (Haircare, stock=1 unit, revenue INR 9,578), and SKU34 (Skincare, stock=1 unit). A stockout on these products directly impacts top-line revenue.

SKU	Product Type	Stock Level (units)	Revenue (INR)	Stockout Risk
SKU31	Skincare	6 units	9,655	CRITICAL
SKU2	Haircare	1 unit	9,578	CRITICAL
SKU34	Skincare	1 unit	9,062	CRITICAL
SKU18	Haircare	10 units	9,365	HIGH
SKU67	Skincare	12 units	9,474	HIGH

5. Profit Margin Analysis: High Margins Overall, but Cosmetics Leads

Despite high defect rates and shipping delays, profit margins are strong across all product types because manufacturing costs (INR 4,726.67 total) and shipping costs (INR 554.81 total) are very low relative to revenue (INR 5,77,604.82). Cosmetics has the highest average profit margin at 99.02%, followed by Skincare at 98.54% and Haircare at 98.41%. Supplier 3 leads on margin efficiency at 99.0% average, while Supplier 4 is the weakest at 98.1%. The primary risk to profitability is not cost, it is quality-driven returns and potential stockout-driven revenue loss.

Supplier	Avg Profit Margin %	Defect Rate %	Delay Rate %	Revenue Contribution %
Supplier 1	98.6%	36.1%	11.1%	27.3%
Supplier 2	98.9%	47.3%	13.6%	21.7%
Supplier 3	99.0%	49.3%	26.7%	16.9%
Supplier 4	98.1%	46.7%	11.1%	15.0%
Supplier 5	98.5%	53.3%	27.8%	19.1%

RECOMMENDATIONS:

1. Recommendation 1:

Problem	Systemic quality failure: 91% of products exceed acceptable defect rate. Supplier 5 at 53.3% and Supplier 3 at 49.3% are the worst offenders. 33 products simultaneously failed inspection AND have high defect rates.
Root Cause	No formal supplier quality management process. Lack of defect thresholds, supplier SLAs, or consequence mechanisms. Cosmetics and Skincare products from Supplier 5 are disproportionately affected.
Action	Immediately audit Supplier 5 and place on a 90-day improvement plan with mandatory defect rate targets (<10%). Introduce supplier scorecards (as modelled in Q45/Q50) as a monthly review mechanism. Consider reducing order volumes from Supplier 5 until defect rates improve. Require pre-shipment quality certificates.
Expected Impact	Reduction of defect-related returns and wastage. If defect rates reduce from 45% to 10%, this protects INR 1,10,343 (Supplier 5 revenue) from being eroded by returns and quality costs. Improved customer satisfaction and brand trust.

2. Recommendation 2:

Problem	53% of products (53 SKUs) have stock below 50 units. 30 of these are high-revenue products above average revenue of INR 5,776. SKU31, SKU2, SKU34 are at 1-6 units effectively out of stock.
Root Cause	No automatic reorder triggers. Inventory planning appears reactive rather than predictive. High-revenue SKUs are not prioritised in restocking cycles. No minimum stock level policy exists.
Action	Implement minimum stock level policy: High Revenue products (>INR 7,000) must maintain minimum 100 units buffer. Set automated reorder triggers at 50 units. Prioritise restocking for the top 30 stockout-risk products identified in Q42. Work with Supplier 1 (best performer, 27 products) to expedite restocking of critical SKUs.
Expected Impact	Prevention of stockout on top-earning SKUs protects an estimated INR 2,00,000+ in revenue from disruption. A 100-unit buffer for high-revenue products represents approximately INR 2-3 lakh in working capital investment against significantly higher potential revenue loss.

3. Recommendation 3:

Problem	Route C has the highest average shipping cost at INR 5.90/unit. Road and Air transportation modes have the worst cost-to-revenue ratio (0.10%). 17% of shipments are delayed despite carriers showing early average delivery.
Root Cause	No route optimisation strategy. High-cost routes are used without comparing carrier alternatives. The 17 delayed shipments likely concentrate on specific route-supplier combinations that have not been identified or addressed.
Action	Conduct route-cost audit: map each delayed shipment to its carrier, route, and supplier to identify the specific bottleneck combinations. Shift eligible shipments from Road/Air to Rail/Sea where feasible (Rail/Sea cost ratio: 0.09% vs 0.10%). Negotiate bulk shipping rates with Carrier A (best average delay performance). Set a target of reducing delay rate from 17% to below 10% within 2 quarters.
Expected Impact	Switching 20% of Road/Air volume to Rail/Sea could reduce shipping costs by approximately 10%. Reducing delay rate to <10% improves customer satisfaction and reduces expedite costs. Estimated annual savings: INR 5,000-8,000 in shipping costs at current scale.

DATA LIMITATIONS:

This analysis was conducted on a single-snapshot dataset of 100 products. The following limitations should be noted when interpreting findings:

- No time dimension: The dataset has no order dates, making trend analysis (seasonal patterns, month-on-month growth) impossible. A date column would significantly enhance demand forecasting.
- No customer-level data: Customer demographics are aggregated categories (Male/Female/Non-binary/Unknown). Individual customer purchase history, churn, or lifetime value cannot be calculated.
- No returns data: While defect rates are high, the dataset does not record actual product returns, warranty claims, or customer complaints, so the financial impact of defects is estimated, not measured.
- Supplier cost structure unknown: Manufacturing costs and shipping costs are at SKU level, but supplier contracts, volume discounts, and penalty clauses are not in the dataset.
- Small sample size: 100 products across 5 suppliers limits statistical confidence. Findings are directionally correct but should be validated with a larger dataset.
- No competitor benchmarking: Revenue and margin comparisons are internal only. Without industry benchmarks, it is unclear whether current performance is above or below market average.

Note: This is a Kaggle practice dataset with synthetically generated defect values. In a real business scenario, defect rates above 10% would trigger immediate supplier termination.

CONCLUSION

The supply chain analysed in this report generates strong top-line revenue of INR 5,77,605 with excellent profit margins averaging 98.5%, indicating that the business model is fundamentally sound. However, two critical operational risks demand immediate attention: the systemic quality failure (average defect rate of 45.54%, industry norm <5%) and the inventory crisis (53% of products on low stock, with the highest-earning SKUs most at risk).

The business should prioritise in this order:

- (1) Supplier quality audit and SLA enforcement, starting with Supplier 5 (53.3% defect rate)
- (2) Emergency inventory replenishment for the 30 high-revenue, low-stock products
- (3) Route and carrier optimisation to reduce the 17% delay rate.

These three actions, if executed within the next quarter, have the potential to protect and grow the INR 5.77 lakh revenue base while significantly reducing quality-related risk exposure.